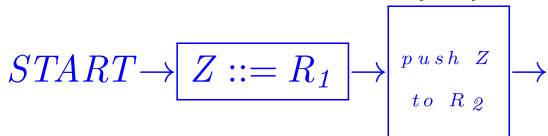


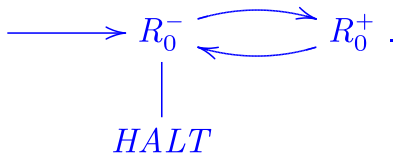
Assume we have a RM H that decides the Halting Problem and derive a contradiction, as follows:

- Let H' be obtained from H by replacing $START \rightarrow$ by



(where Z is a register not mentioned in H 's program).

- Let C be obtained from H' by replacing each $HALT$ (& each erroneous halt) by



- Let $c \in \mathbb{N}$ be the index of C